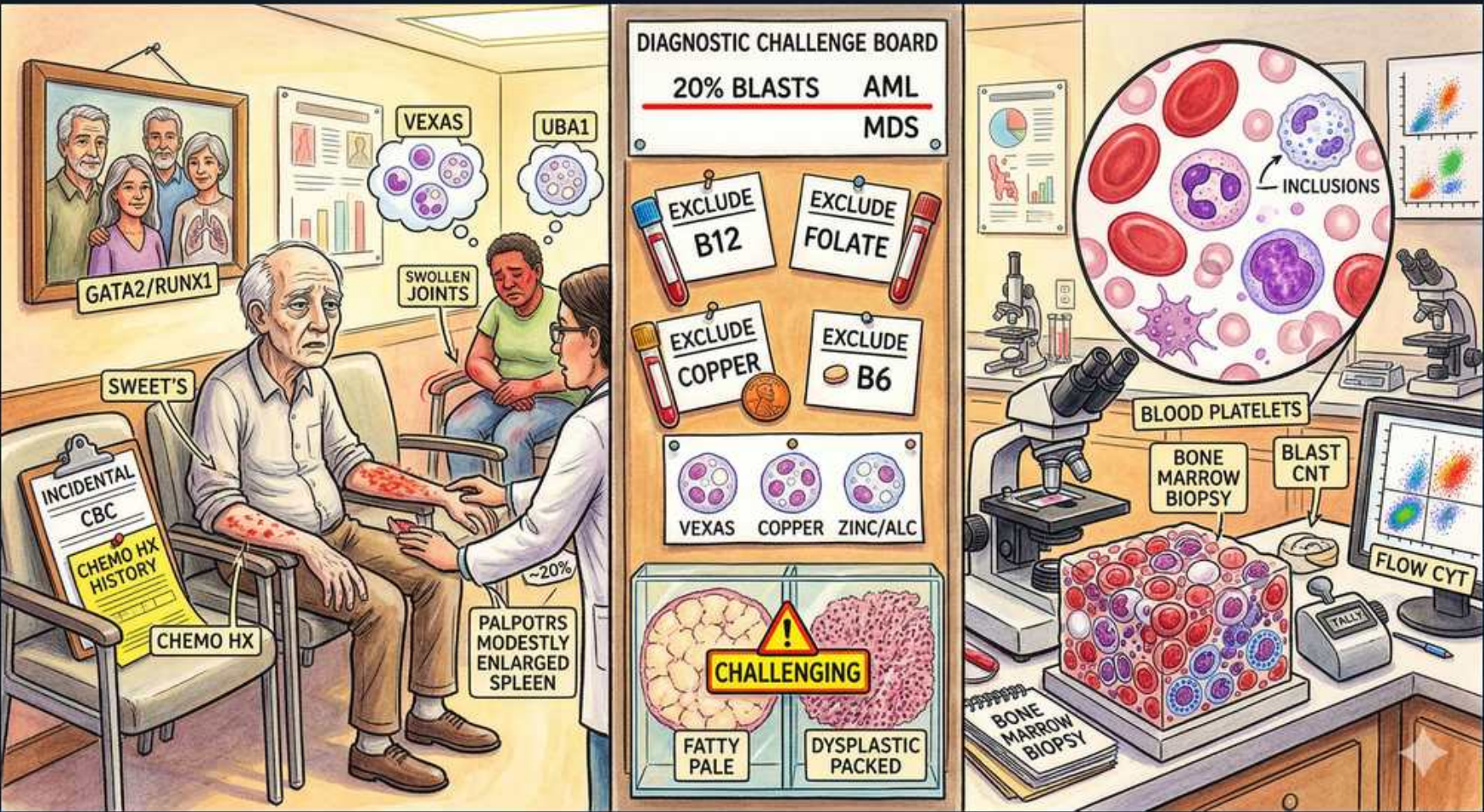


MDS — The MDS Clinic (Presentation, Workup & Exclusions)



1. Elderly patients

WHAT YOU SEE

Framed photos of grey-haired older adults hung on the clinic wall — MDS is fundamentally a disease of aging marrow, so the gallery of elderly faces IS the typical patient.

WHAT IT ENCODES

MDS is fundamentally a disease of aging hematopoiesis — median age at diagnosis ~70, incidence rising sharply after 60. The typical presentation is an asymptomatic or fatigued older adult with unexplained, persistent cytopenia(s), most often a macrocytic anemia with a low reticulocyte count. Always anchor the differential for chronic, unexplained cytopenia in an elderly patient on MDS once reversible causes are excluded.

BOARD TRAP

Elderly patient, MCV 105, persistent anemia, normal B12/folate/TSH. WRONG answer: settle on 'anemia of chronic disease' or aging. Discriminator: ACD is typically normocytic/microcytic with high ferritin and low TIBC; a persistent macrocytic, hypoproliferative cytopenia in the elderly after excluding B12/folate/thyroid/drugs warrants marrow evaluation for MDS.

2. VEXAS

WHAT YOU SEE

A wall banner reading VEXAS beside inflamed-looking cells with cytoplasmic vacuoles — the vacuoles are the visual signature of the VEXAS syndrome.

WHAT IT ENCODES

VEXAS (Vacuoles, E1 enzyme, X-linked, Autoinflammatory, Somatic) is a recently described syndrome from somatic UBA1 mutations in myeloid precursors. It affects older men with treatment-refractory inflammation (fevers, chondritis, vasculitis, skin lesions, VTE) plus macrocytic anemia and cytoplasmic vacuoles in myeloid/erythroid precursors; ~50% have or progress to MDS. It links autoinflammation and clonal myeloid disease.

BOARD TRAP

Older man with relapsing polychondritis, fevers, macrocytic anemia not responding to steroids; ANCA negative. WRONG answer: chase another classic vasculitis/connective-tissue disease. Discriminator: precursor vacuoles + macrocytic anemia + steroid-refractory autoinflammation in an older male = test UBA1 for VEXAS rather than continuing empiric rheumatology workups.

3. UBA1

WHAT YOU SEE

A speech-bubble labelled UBA1 — the single gene whose somatic mutation drives VEXAS.

WHAT IT ENCODES

UBA1 encodes the E1 ubiquitin-activating enzyme; a somatic UBA1 mutation (commonly at p.Met41) in hematopoietic precursors is the defining driver of VEXAS. The defective ubiquitylation triggers innate-immune/autoinflammatory activation while seeding clonal, dysplastic myelopoiesis — mechanistically tying systemic inflammation to MDS-like marrow changes.

BOARD TRAP

A man with VEXAS phenotype but the lab ordered only germline genetics. WRONG answer: rule out VEXAS because no germline mutation found. Discriminator: UBA1 in VEXAS is a SOMATIC mutation restricted to myeloid cells — test peripheral blood/marrow myeloid DNA, not germline (e.g., skin fibroblasts), or you will miss the diagnosis.

4. GATA2 / RUNX1

WHAT YOU SEE

A GATA2/RUNX1 tag pinned to family portraits — the family photos signal these are inherited, familial MDS-predisposition genes.

WHAT IT ENCODES

Germline predisposition syndromes cause familial MDS/AML. GATA2 deficiency: monocytopenia, B/NK-cell lymphopenia, disseminated mycobacterial/HPV infections, lymphedema (Emberger), pulmonary alveolar proteinosis, with MDS in young adults. RUNX1 germline: familial platelet disorder with thrombocytopenia/bleeding and lifelong MDS/AML risk. Suspect these in young MDS or a positive family history — and screen related stem-cell donors.

BOARD TRAP

A 24-year-old with MDS; team assumes it must be therapy-related. WRONG answer: label it secondary/therapy-related without exposure history. Discriminator: young-onset MDS with no chemo/radiation history, plus features like familial thrombocytopenia, recurrent atypical infections, or lymphedema, points to inherited GATA2/RUNX1 predisposition — which also mandates germline testing of a sibling before using them as a transplant donor.

5. Sweet's

WHAT YOU SEE

A 'Sweet's' tag pointing to tender red raised plaques on the patient's skin — the angry skin plaques ARE Sweet syndrome, a paraneoplastic marker of myeloid disease.

WHAT IT ENCODES

Sweet syndrome (acute febrile neutrophilic dermatosis) — tender, edematous, red plaques with fever, neutrophilia, and dense dermal neutrophil infiltrate — is a paraneoplastic association of myeloid malignancy, especially MDS/AML. Other immune phenomena (relapsing polychondritis, vasculitis, neutrophilic dermatoses) can herald or accompany MDS. It responds to corticosteroids but should prompt evaluation of the underlying clone.

BOARD TRAP

Recurrent Sweet syndrome in an older adult treated with steroids alone. WRONG answer: idiopathic dermatosis, no further workup. Discriminator: malignancy-associated Sweet syndrome (often atypical/bullous, recurrent, cytopenic rather than neutrophilic) should trigger a CBC and marrow to exclude underlying MDS/AML.

6. Swollen joints

WHAT YOU SEE

The seated patient's hands/joints drawn red, puffy and inflamed — the visibly swollen joints depict the paraneoplastic inflammatory arthritis of MDS/VEXAS.

WHAT IT ENCODES

MDS (and especially VEXAS) can present with systemic inflammatory/rheumatologic manifestations — inflammatory arthritis, polychondritis, vasculitis, and seronegative arthropathies — as paraneoplastic immune dysregulation rather than a primary rheumatic disease. These can predate the hematologic diagnosis and are frequently refractory to standard immunosuppression.

BOARD TRAP

Older patient with seronegative inflammatory arthritis, cytopenias, and constitutional symptoms not responding to DMARDs. WRONG answer: escalate rheumatologic therapy indefinitely. Discriminator: refractory 'rheumatic' disease in an older adult with unexplained cytopenias should prompt hematologic evaluation (CBC, marrow, UBA1) for MDS/VEXAS as the true driver.

7. Incidental CBC

WHAT YOU SEE

A CBC lab printout stamped 'INCIDENTAL' on the chart — the flagged routine blood report shows MDS is most often caught by chance on a screening CBC.

WHAT IT ENCODES

MDS is very commonly discovered INCIDENTALLY on a routine CBC: an isolated macrocytic anemia, neutropenia, thrombocytopenia, or monocytosis in an otherwise well older adult. The macrocytosis is hypoproliferative (low/inappropriately normal reticulocyte response), distinguishing it from hemolysis. Persistent unexplained cytopenia (lasting months) is the entry point to the MDS workup.

BOARD TRAP

Incidental MCV 108 with mild anemia, normal B12/folate; clinician repeats the CBC indefinitely without action. WRONG answer: keep watching a persistent macrocytic anemia without explanation. Discriminator: persistent unexplained macrocytosis/cytopenia after excluding B12, folate, thyroid, alcohol, and drugs is itself an indication for marrow examination, not indefinite observation.

8. Chemo Hx

WHAT YOU SEE

A chart entry labelled 'CHEMO HX' — the documented prior-chemotherapy history is the clue to therapy-related MDS.

WHAT IT ENCODES

A history of cytotoxic therapy is pivotal: alkylating agents/radiation cause therapy-related MDS at ~5–7 years (with -5/-7, complex karyotype, TP53), while topoisomerase-II inhibitors cause it at ~1–3 years (with 11q23/KMT2A). Therapy-related MDS is a distinct, poor-prognosis category with high AML transformation risk and chemo-resistance. Always document prior cancer treatment in any new cytopenia.

BOARD TRAP

New pancytopenia 6 years after CHOP + radiation for lymphoma; called 'marrow suppression from old chemo.' WRONG answer: dismiss as benign late myelosuppression. Discriminator: cytopenias years after alkylator/radiation with dysplasia and adverse cytogenetics = therapy-related MDS, which is prognostically worse than de novo and should be staged and referred for transplant, not observed.

9. Modest splenomegaly

WHAT YOU SEE

A clinical note reading the spleen is only MODESTLY enlarged — the just-slightly-big spleen is the bedside clue that this is pure MDS, not an MDS/MPN overlap.

WHAT IT ENCODES

In pure MDS the spleen is usually normal or only modestly enlarged. Prominent splenomegaly, leukocytosis with monocytosis, or marrow fibrosis suggests an MDS/MPN overlap — chronic myelomonocytic leukemia (CMML), defined by persistent monocytosis $\geq 1 \times 10^9/L$ ($\geq 10\%$ of WBC) — or primary myelofibrosis. The degree of splenomegaly is a useful bedside discriminator.

BOARD TRAP

Older patient with anemia, dysplasia, monocytosis, and a large spleen; labeled standard MDS. WRONG answer: classify as ordinary MDS. Discriminator: persistent monocytosis $\geq 1 \times 10^9/L$ plus splenomegaly defines CMML (MDS/MPN overlap), not pure MDS; massive splenomegaly should also raise myelofibrosis.

10. 20% blasts → AML

WHAT YOU SEE

The central 'DIAGNOSTIC CHALLENGE BOARD' headed '20% BLASTS' with arrows splitting toward AML vs MDS — the 20% blast line on the board IS the threshold dividing the two diagnoses.

WHAT IT ENCODES

The blast count separates MDS from AML: MDS requires <20% myeloblasts in blood and marrow; $\geq 20\%$ defines AML. MDS-with-excess-blasts is subdivided (IB1 5–9%, IB2 10–19% marrow blasts). Auer rods or AML-defining genetics (t(8;21), inv(16), PML-RARA, KMT2A) can establish AML even below 20% blasts.

BOARD TRAP

Marrow with 22% blasts described as 'very high-risk MDS.' WRONG answer: keep calling it MDS and start an HMA for low-risk disease. Discriminator: $\geq 20\%$ blasts (or an AML-defining genetic lesion at any count) IS AML, which mandates AML-directed therapy/induction or HMA+venetoclax — not the MDS supportive-care pathway.

11. Exclude B12 / folate

WHAT YOU SEE

Two blood tubes on the board stamped 'EXCLUDE B12' and 'EXCLUDE FOLATE' — the crossed-out vitamin tubes mean these reversible mimics must be ruled out before calling it MDS.

WHAT IT ENCODES

B12 and folate deficiency must be excluded before diagnosing MDS because megaloblastic anemia produces near-identical findings: macro-ovalocytosis, hypersegmented neutrophils, megaloblastoid erythroids, and even pancytopenia with a hypercellular marrow. Check serum B12 and folate; if borderline, confirm with methylmalonic acid ($\uparrow$ in B12 deficiency) and homocysteine. These are reversible with repletion.

BOARD TRAP

Pancytopenia, MCV 112, hypercellular marrow with dysplasia; marrow read as MDS before nutritional labs return. WRONG answer: diagnose and treat MDS immediately. Discriminator: B12/folate deficiency mimics MDS morphology exactly — replete and recheck the CBC/blood film first; a brisk reticulocyte and count recovery proves it was nutritional, not clonal MDS.

12. Exclude copper / zinc / B6

WHAT YOU SEE

Board cards reading 'EXCLUDE COPPER' and 'EXCLUDE B6' beside copper/zinc/alcohol cell icons — the eliminated deficiency cards are reversible causes of dysplasia/sideroblasts to exclude first.

WHAT IT ENCODES

Several reversible deficiencies/toxins cause secondary dysplasia and sideroblasts. Copper deficiency (from zinc excess — denture cream, supplements, bariatric surgery) causes anemia + neutropenia with ring sideroblasts and vacuolated precursors, often with a myeloneuropathy. B6 (pyridoxine) deficiency, alcohol, lead, and isoniazid/linezolid also produce sideroblastic/dysplastic changes. Check copper, zinc, ceruloplasmin, B6, and a careful drug/alcohol history.

BOARD TRAP

Anemia + neutropenia with ring sideroblasts and a spastic gait, recent gastric bypass. WRONG answer: diagnose MDS-RS (ring-sideroblast MDS). Discriminator: ring sideroblasts are NOT specific for MDS — copper deficiency (often zinc-induced, post-bariatric, with neuropathy), alcohol, lead, and drugs (linezolid, INH) must be excluded; copper repletion reverses the cytopenias.

13. Bone marrow biopsy

WHAT YOU SEE

A bone-marrow biopsy core drawn densely packed with cells (hypercellular) — the overcrowded marrow core IS the MDS paradox: a full marrow despite low blood counts.

WHAT IT ENCODES

Diagnosis requires marrow aspirate + biopsy with cytogenetics and increasingly NGS. Classic findings: a HYPERcellular (occasionally normocellular) marrow with morphologic dysplasia in ≥10% of cells of at least one lineage, plus a blast count <20%, despite peripheral cytopenia(s) — the hallmark ineffective hematopoiesis. An iron stain quantifies ring sideroblasts; cytogenetics/FISH and IPSS-R risk follow.

BOARD TRAP

Peripheral cytopenia with a packed hypercellular marrow; clinician expects an empty marrow to confirm a 'failure' state. WRONG answer: a cellular marrow rules out a bone-marrow failure disorder. Discriminator: the MDS PARADOX is cytopenia WITH a hypercellular marrow (cells die before release) — a hypercellular marrow supports MDS, whereas an empty/fatty marrow points to aplastic anemia.

14. Dysplastic vs fatty marrow

WHAT YOU SEE

Side-by-side marrow panels — a pale, empty 'FATTY' marrow versus a 'DYSPLASTIC PACKED' crowded marrow — the contrast shows hypocellular (aplastic) vs hypercellular dysplastic (MDS).

WHAT IT ENCODES

Marrow cellularity discriminates entities: typical MDS is hypercellular and dysplastic, whereas a hypocellular/fatty marrow (cellularity well below age-adjusted norm, ~100–age%) suggests aplastic anemia. Hypoplastic MDS is the rarer overlap variant — a hypocellular marrow that nonetheless shows definite dysplasia and/or clonal cytogenetics.

BOARD TRAP

Pancytopenia, hypocellular fatty marrow, no clear dysplasia, normal cytogenetics in a young patient. WRONG answer: diagnose hypoplastic MDS. Discriminator: a fatty, hypocellular marrow WITHOUT convincing dysplasia or clonal markers favors aplastic anemia; only definite dysplasia or clonal cytogenetics reclassifies it as hypoplastic MDS.

15. Dysplastic inclusions

WHAT YOU SEE

A microscope blow-up of cells with abnormal nuclei and inclusions (top-right 'INCLUSIONS') — the misshapen cells under magnification ARE the dysplasia that defines MDS.

WHAT IT ENCODES

Morphologic dysplasia is the diagnostic hallmark across lineages: granulocytic — hypogranular, hyposegmented pseudo-Pelger-Huët (bilobed 'pince-nez') neutrophils; erythroid — megaloblastoid maturation, nuclear budding/irregularity, ring sideroblasts on iron stain; megakaryocytic — small mono-/hypolobated micromegakaryocytes. Dysplasia must involve ≥10% of cells in a lineage to count toward diagnosis.

BOARD TRAP

Hypogranular bilobed neutrophils called a left shift or artifact. WRONG answer: dismiss pseudo-Pelger-Huët cells as benign. Discriminator: acquired pseudo-Pelger-Huët (hypolobated, hypogranular neutrophils) is a key dysplastic finding of MDS — distinguish from the benign inherited Pelger-Huët anomaly and from a reactive left shift by its association with other dysplasia and cytopenias.

16. Flow cytometry

WHAT YOU SEE

Flow-cytometry dot-plots beside a 'BLAST CNT' panel on the right — the scatter plots represent flow quantifying CD34+ blasts and aberrant maturation.

WHAT IT ENCODES

Flow cytometry supports diagnosis and risk by quantifying blasts (CD34+ progenitors) and detecting aberrant maturation/antigen expression (e.g., abnormal CD13/CD16 granulocyte patterns, aberrant CD7/CD56). It cannot replace morphology and cytogenetics for diagnosis, but the blast percentage it helps establish feeds directly into IPSS-R/IPSS-M prognostic scoring.

BOARD TRAP

Flow shows 4% CD34+ blasts and the team concludes the marrow blast count is definitively 4% for staging. WRONG answer: use flow blast % as the official count. Discriminator: the diagnostic/prognostic blast percentage in MDS is determined by MORPHOLOGY on a 500-cell aspirate differential — flow can under- or over-estimate blasts (dilution, gating), so it complements but does not substitute for the manual count.